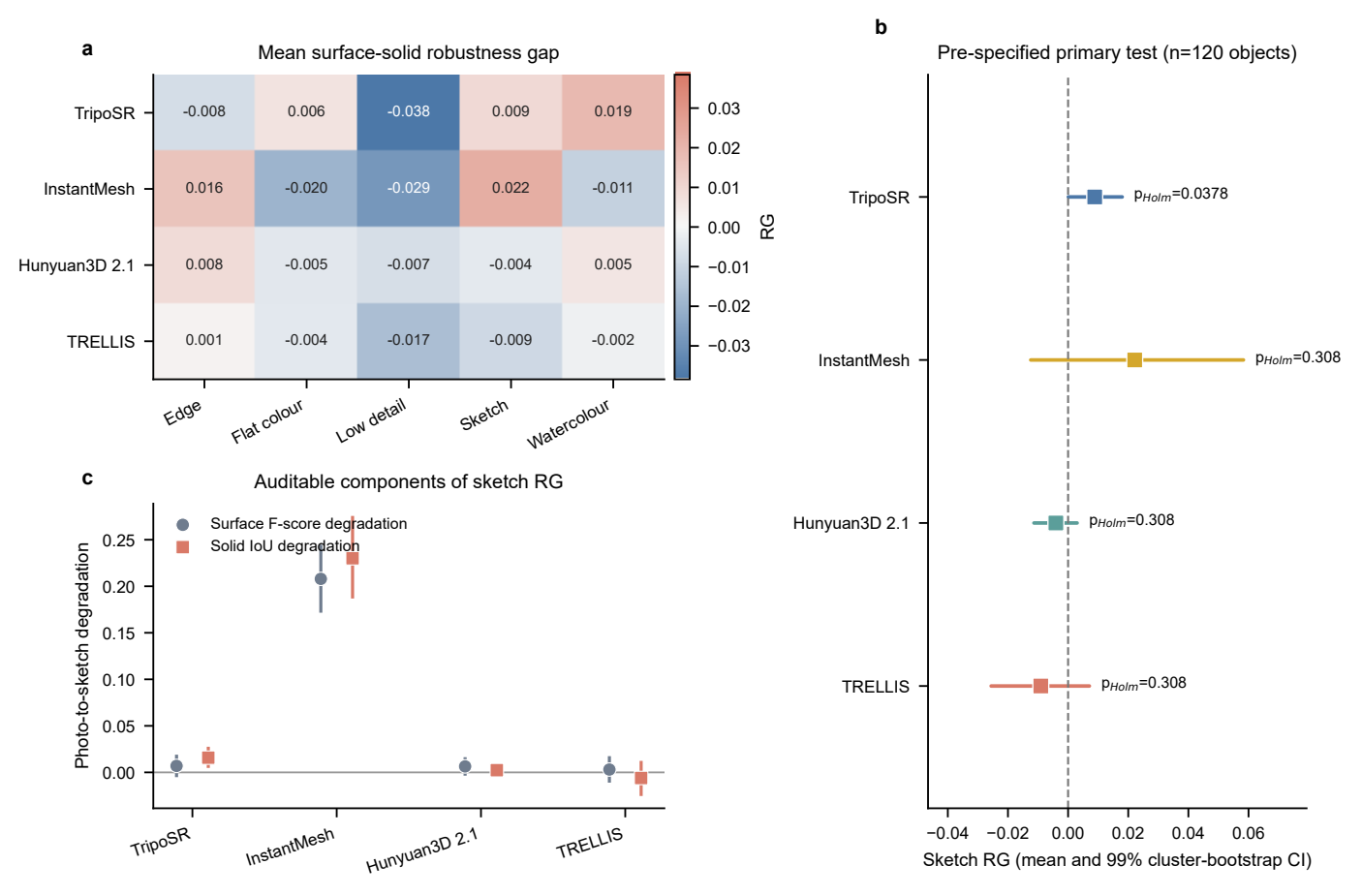

a
Mean surface-solid robustness gap
TripoSR
InstantMesh
Hunyuan3D 2.1
TRELLIS
-0.008
0.006
-0.038
0.009
0.019
0.016
-0.020
-0.029
0.022
-0.011
0.008
-0.005
-0.007
-0.004
0.005
0.001
-0.004
-0.017
-0.009
-0.002
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
RG
Edge
Flat colour
Low detail
Sketch
Watercolour
b
Pre-specified primary test (n=120 objects)
TripoSR
InstantMesh
Hunyuan3D 2.1
TRELLIS
pHolm=0.0378
pHolm=0.308
pHolm=0.308
pHolm=0.308
−0.04
−0.02
0.00
0.02
0.04
0.06
Sketch RG (mean and 99% cluster-bootstrap CI)
c
Auditable components of sketch RG
Surface F-score degradation
Solid IoU degradation
Photo-to-sketch degradation
0.25
0.20
0.15
0.10
0.05
0.00
TripoSR
InstantMesh
Hunyuan3D 2.1
TRELLIS